

# Joe Pan — Senior iOS Engineer

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## SUMMARY

Over **15 years** of hands-on iOS development experience across a wide range of app types — MFi hardware accessories, instant messaging, video streaming, and fintech — participating in every stage of the platform's evolution from pure Objective-C to modern Swift and SwiftUI.

Focused on **architecture design**, with a core conviction that architecture should be grounded in Apple's own frameworks — studying the design intent behind UIKit, SwiftUI, and Swift Concurrency, then building loosely coupled, maintainable solutions from that foundation rather than adopting third-party frameworks that diverge from the platform's native model. This **native-first** philosophy extends to dependency management: Apple's native APIs are always the first choice, keeping maintenance overhead and binary size in check. Adapts quickly to existing team conventions and designs custom solutions when established patterns fall short; MVVMC is a direct expression of this conviction.

Years of hands-on experience across nearly every common app feature type means problems are diagnosed fast — most scenarios have been encountered before, making it possible to zero in on the key information and reach a viable solution quickly.

## SKILLS

### Languages

Swift 3.0 – present Objective-C 2009 – present

### UI

Proficient in both UIKit and SwiftUI, capable of building complex custom interfaces from scratch. The two are combined as the core foundation of the MVVMC architecture, with `UIHostingController` as the navigation unit and SwiftUI handling rendering. Early study of cocos2d for iPhone established a solid 2D graphics foundation, which carried over into deep familiarity with SpriteKit — used as a fallback for animations and interactions that UIKit and SwiftUI cannot achieve natively.

UIKit SwiftUI UIHostingController hybrid AutoLayout Core Animation SpriteKit

### Architecture

Not tied to a single pattern — prioritises maintainability and team adaptability. Has led or contributed to the adoption and refactoring of multiple architectures; designs custom solutions when standard ones prove inadequate.

MVC MVVM Clean Swift TCA MVVMC (custom) @Observable / @MainActor

### Swift Concurrency

Migrated an existing Swift 5+ codebase to full Swift 6 readiness through systematic adoption of Swift Concurrency.

async/await @MainActor actor Sendable Task / TaskGroup

### Networking

URLSession is the default (native-first); gRPC is used where the project requires it. Practical experience with Alamofire and other third-party networking libraries.

URLSession gRPC (grpc-swift / connect-swift) Alamofire

### Real-Time Communication

Hands-on production experience with all major messaging protocols, covering both text-based communication and live video streaming.

WebSocket XMPP MQTT SignalR VoIP Live Streaming (HaishinKit / ijkplayer / ReplayKit)

### Persistence

Native solutions are preferred; major third-party options have also been used in production.

CoreData SwiftData Keychain NSKeyedArchiver Realm fmdb

### Bluetooth

CoreBluetooth iBeacon

### Toolchain

| Category           | Tools                                                           |
|--------------------|-----------------------------------------------------------------|
| Version Control    | Git                                                             |
| Package Management | Swift Package Manager, CocoaPods, Carthage                      |
| Project Management | XcodeGen, Tuist                                                 |
| CI / CD            | Xcode Cloud, Jenkins, Fastlane, Firebase App Distribution       |
| AI Development     | Claude (CLAUDE.md + Skill specs for systematic code generation) |

## EXPERIENCE

### DRACO EVOLUTION

Apr 2025 – Jun 2026

Senior iOS Engineer · Took over from an outsourced team; sole iOS engineer at the company

Independently assessed code quality, paid down technical debt, and led the technical architecture evolution.

- TCA upgrade and removal:** Inherited the project at TCA 1.15.2 and upgraded incrementally to 1.22.2 (1.18.0 → 1.19.1 → 1.20.2 → 1.21.1 → 1.22.1 → 1.22.2). During routine maintenance and feature development, identified that TCA's built-in Navigation Stack could not accommodate the app's non-standard routing requirements, prompting a gradual architectural rewrite that went live on 18 Sep 2025.
- Custom MVVMC architecture (GitHub):** After removing TCA, found that SwiftUI's native NavigationStack still couldn't handle complex non-standard routing (e.g. simultaneously dismissing a checkout flow, switching tabs, and clearing a navigation stack). Combined with a deliberate decision not to base the entire codebase on a third-party architecture framework, drew on concepts from TCA and MVI to design MVVMC from scratch in native Swift — extending MVVM with a HostController layer (C): M handles State / Domain Models / DTOs; VM uses `@Observable @MainActor` with a single entry point (`doAction`) for business logic; V is pure SwiftUI with zero navigation or business logic; C (`UIHostingController`) is the sole Router with all navigation centralised through `AppRouter.shared`. Open-sourced with an MCP Server so Claude Code can access the architecture spec from any project.
- Swift 5 → Ready for Swift 6 migration:** Inherited a Swift 5+ codebase and systematically introduced `async/await`, `@MainActor`, `actor`, and `Sendable`, completing the Swift 6 compatibility migration in parallel with the TCA removal and MVVMC refactor.
- Xcode Cloud CI/CD pipeline:** Dev builds distribute automatically to TestFlight internal testers; production builds distribute to TestFlight external testers — fully automated, no manual steps required.
- US App Store release:** Published exclusively on the US App Store to comply with SEC regulations; subsequently taken down following an API shutdown.
- AI-assisted development workflow (Claude + Skill):** Authored `CLAUDE.md` and a suite of Skill documents (SwiftUI, ViewModel, HostController, Model, Swift Concurrency, etc.) so that Claude generates feature code that conforms to the project's existing architecture, significantly reducing time-to-feature.

### Everictory Technology

Sep 2023 – Sep 2024

iOS Engineer · Lead developer in a 3-person iOS team; the other two focused on an in-house K-Line library and a separate VPN app

Primary maintainer of a fintech trading platform and [JinTian GT](#) (a precious-metals trading app), plus contributions to an in-house IM client built on Tinode.

#### Fintech Trading App (demo)

- Introduced XcodeGen to eliminate persistent `.xcodeproj` merge conflicts in a multi-developer workflow.
- Integrated SignalR for real-time quote streaming; resolved UICollectionView cell-update jank during scroll using `CADisplayLink + RunLoop`.
- Replaced UITableView with UICollectionView across the codebase to enable richer layouts and more flexible cell composition; maintained and extended the in-house K-Line charting library.
- Wrote shell scripts to switch between dev / QAT / production environments; implemented in-app live language switching.

#### JinTian GT App (demo)

- Swift / Objective-C mixed-language development; integrated gRPC for data transport.
- Applied the [Rate Limiting UITableView/UICollectionView Reloads](#) technique to throttle excessive reloads triggered by high-frequency gRPC updates.
- Implemented a review-mode UI-hiding mechanism to pass App Store review.
- Built a CI/CD pipeline with Jenkins + Fastlane; used Firebase App Distribution for dev / QAT builds.
- Adopted Xcode 15 asset symbol generation; built a JavaScript bridge for bidirectional HTML5 ↔ native communication.

#### IM Client (Tinode-based)

- Removed Storyboard in favour of fully programmatic UI to enable deep customisation of Tinode's default interface.
- With no Golang engineer on the team, self-taught Go and read through the Tinode server codebase to understand server behaviour and accurately define the boundaries of IM feature development.

### Heng Yuan Technology Co., Ltd.

Apr 2022 – Jul 2023

iOS Engineer · Sole iOS engineer on the project (demo)

Took over an Objective-C codebase and led the technical modernisation of an app covering short video, long video, comics, novels, games, live streaming, and chat.

- Led ObjC → Swift migration:** Defined a gradual strategy — all new features written in Swift, legacy modules refactored incrementally.
- Introduced XcodeGen to eliminate Git project-file conflicts; researched Tuist and evaluated both tools.
- Refactored local HLS playback architecture:** The original approach downloaded m3u8 files locally and served them through GCDWebServer — high maintenance cost. Replaced with a custom `AVAssetResourceLoaderDelegate`: substitutes `file://` with a custom `local://` scheme so AVPlayer delegates the request to the loader, which swaps back to `file://` and reads the local data. Removed the GCDWebServer dependency and reduced binary size ([reference article](#)).
- Built comic reader, novel reader, and embedded game pages — all using WebView with bidirectional native communication.
- Integrated a live-streaming SDK written in Flutter, resolving Flutter module embedding issues and debugging across framework boundaries.
- Coordinated with a third-party signing vendor for enterprise distribution outside the App Store.

### Gamania Group

Jun 2019 – Feb 2022

iOS Contract Engineer

#### BeanFun App

- Collaborated with the team to introduce Clean Swift architecture, progressively migrating Objective-C modules to Swift.
- BeanFun loads the majority of its features in WebViews and requires hot-reload support; implemented deep WebView ↔ native bidirectional communication covering JavaScript bridge design, event propagation, and page state synchronisation.

### Gamania Group

Sep 2018 – Mar 2019

iOS Contract Engineer · Lead iOS developer

#### In-House Enterprise App "teamup!"

Internal project management and approval platform supporting instant messaging, digital sign-off, and access control. Used by Gamania's CEO during an Antarctic expedition as a remote management tool ([press coverage](#)).

- Optimised for low-bandwidth conditions: cached API responses with fmdb for offline use; implemented multi-tier image caching (thumbnail-first display, background download and swap to full resolution) to keep the app functional on slow connections.

### Wistron ITS

Aug 2016 – Mar 2018

iOS Engineer · On-site at Cathay Life Insurance

Primary maintainer of the [Cathay Life Insurance](#) app. The codebase was Objective-C; following Cathay's internal decision to adopt Swift, took responsibility for implementing new features in Swift and progressively migrating legacy Objective-C modules.

### Hoter Information

Mar 2016 – Jun 2016

iOS Engineer

Developed an in-house hotel workforce management app for staff scheduling, real-time reporting, and check-in. The backend chose SignalR for the chat feature — an uncommon choice at a time when XMPP and MQTT dominated the space — providing first-hand experience with the SignalR protocol. Also implemented geofenced check-in using Core Location.

### Internet Mobile Technology

Apr 2015 – Jan 2016

iOS Engineer

Built a shopping app centred on a gamification concept inspired by Tap Titans: users earn virtual currency or discount coupons through in-app gameplay, redeemable in the store. Implemented chat using XMPP and built the game module with SpriteKit — the first experience integrating a game engine into an e-commerce app.

### Earlier Experience (2010 – 2015)

| Company                     | Period              | Key Work                                                                                                                                                                                         |
|-----------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PiPiMy                      | Jan 2015 – Mar 2015 | C2C second-hand trading app (Beta)                                                                                                                                                               |
| Timeline Technology         | May 2012 – Jun 2014 | Template-based storefront app framework / <a href="#">coupon app</a>                                                                                                                             |
| JamZoo                      | Nov 2012 – Jun 2013 | Migrated "Single Bank" app chat from timer polling to MQTT (following Facebook's adoption of MQTT), resolving lag; built <a href="#">offline-capable car rental app</a> and HTML5 e-book WebView |
| Viamedia Mobile Corporation | Sep 2010 – Feb 2012 | <a href="#">Media Player interactive app</a> , <a href="#">mini-games</a> , BLE                                                                                                                  |
| Sunplus Semiconductor       | Mar 2010 – Jul 2010 | <a href="#">MFi hardware accessory app</a> , internet radio, local music playback                                                                                                                |

## PROJECTS

### MVVMC

GitHub

A four-layer iOS architecture (M / VM / V / C) designed to overcome the limitations of SwiftUI's native navigation. Uses `UIHostingController` as the navigation unit and centralises all routing through `AppRouter.shared`, keeping SwiftUI views free of any navigation dependencies. Includes a runnable Demo project and an MCP Server for Claude Code integration.

### WebParser

GitHub

A Swift Package for web parsing built on WKWebView off-screen rendering, designed for dynamic pages that require JavaScript execution (SPAs, dynamic DOM). Exposes a type-safe generic Mapper interface (JSON → `Decodable`, custom Regex extraction), fully Swift 6 strict concurrency compliant, tested with Swift Testing, distributed via SPM with GitHub Actions CI.

## EDUCATION

| Institution                        | Programme                                  | Period              |
|------------------------------------|--------------------------------------------|---------------------|
| Chaoyang University of Technology  | Information Management (evening programme) | Sep 2004 – Jan 2007 |
| Kao Yuan University (5-year)       | Electronic Engineering                     | —                   |
| Institute for Information Industry | iOS Development (6-month programme)        | 2009                |

*Graduation project: developed rhythm game Block Ten using cocos2d for iPhone, winning Gold Award at the 2009 Digital Content Creation Competition (Mobile Game category). After graduating, contracted with classmates to build a music app.*

**Languages:** Mandarin Chinese (native) English (technical reading and writing)